

Christian Döring

Graduate Student

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Education

2026 – present
September

EDIC PhD Fellow,

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

2023 – 2026
April March

M.Sc. Electrical and Computer Engineering,

Technical University of Munich

Thesis Title: Representing Mesoscale Detail in Neural Materials

2019 – 2023
October March

B.Sc. Electrical and Computer Engineering,

Technical University of Munich

Thesis Title: Evaluation of Differentiable Inverse Rendering using Multi-View RGB Data

2011 – 2019
September June

Abitur (A-Levels), *Gymnasium Bruckmühl*

Publications

2026

Real-time Rendering with a Neural Irradiance Volume

Arno Coomans, Giacomo Nazzaro, Edoardo A. Dominici, Christian Döring, Floor Verhoeven, Konstantinos Vardis, Markus Steinberger

📄 arXiv

2024

Real-time Neural Rendering of Dynamic Light Fields

Arno Coomans, Edoardo A. Dominici, Christian Döring, Joerg H. Mueller, Jozef Hladky, Markus Steinberger

📄 Project

📄 In Computer Graphics Forum (EG), 2024

Experience

2026 – 2026
June Aug

Research Intern, *Disney Research Zurich,*

○ Post render edits

2026 – 2026
May May

Research Intern, *Realistic Graphics Laboratory, EPFL*

○ Neural Materials

2025 – 2026
September March

Masters Thesis, *Realistic Graphics Laboratory, EPFL*

○ Representing Mesoscale Detail in Neural Materials

2025 – 2025
April August

Research Intern, *NVIDIA Zurich*

○ Function Level Caching for Dr.Jit and Mitsuba 3

○ Hash Grid for Dr.Jit

○ Differentiable Radio Frequency Modeling with Sionna RT

2024 – 2025
April April

Research Working Student, *Huawei Technologies* Munich

- Development on Dr.Jit/Mitsuba 3
- Real-time Neural Rendering Research

2023 – 2024
August February

Research Intern, *Huawei Technologies* Munich

- Real-time Neural Rendering Research

2021 – 2021
July August

Embedded Systems Intern, *Aurum GmbH* Munich

- Developed NFC library for STM32 in C

2017
July

Intern, *Lauterbach GmbH*

2017
July

Intern, *Electronic Theater Controls (ETC)* Holzkirchen

Side Projects

Hephaestus, Just In Time Compiler (JIT) for Vulkan, inspired by Dr.Jit. Implemented with own render graph solution. Includes cooperative matrix multiplication (KHR) and a port of tiny-cuda-nn in GLSL.

◆ Source

Vulkan Path Tracer, Path tracer written in Rust using the screen-13 library. It supports the Disney BSDF with Next Event Estimation.

◆ Source

Mitsuba 3 Experiments, Implementation of forward and differentiable path tracing algorithms in Mitsuba 3, such as ReSTIR GI and Large Steps in Inverse Rendering

Skills

Programming:

- C/C++, Rust
- CUDA, Vulkan
- Python, Lua
- LaTeX, Typst

Languages:

- **German** (native)
- English (B2+/C1)